

Descriptive Statistics

Rex Shao-Yu Wang

Department of Agronomy, National Taiwan University

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Outline

Import Data

Descriptive Statistics

Practice

Import Data

Usually, you won't type in data manually; you'll get it from a file

```
# Load a package
install.package("tidyverse")
library(tidyverse)

# Read a csv file
read_csv("file.csv")

# Read a txt file
read_delim("file.txt")

# Read an excel file
read_excel("file.xlsx")
```

Example: iris data set

```
iris_data <- read_csv("file_path")
```

How to Find the File's Path

Mac OS

Find your file → **Right-click** → Press Option key → Select **“Copy as Pathname”**

Windows

Find your file → **Right-click** → Select **“Copy as path”**

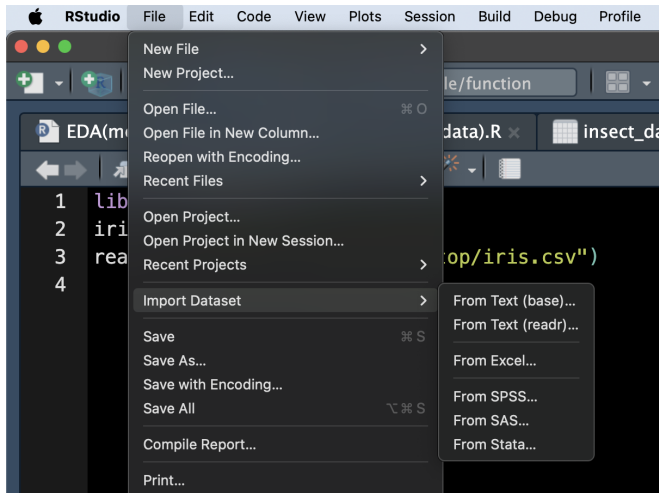
(Note: You may need to click “Show more options” first on Win11)

Pro Tip

In R, remember to change backslashes \ to forward slashes / or use double backslashes \\ for Windows paths!

Other way to import data (User Friendly)

File → Import Dataset → From Text (readr)



Other way to import data

Click Browser to choose your data → Click Import to put your data into R environment

Import Text Data

1. Click Browse to choose your data

File/URL:

Browse...

Data Preview:

Import Options:

Name:

☒ First Row as Names

Delimiter:

Escape:

Skip:

☒ Trim Spaces

Quotes:

Comment:

☒ Open Data Viewer

Locale:

NA:

Code Preview:

```
library(readr)
dataset <- read_csv(NULL)
View(dataset)
```

Reading rectangular data using readr

2. Click Import to put your data into R environment

Import

Cancel

Check the iris dataset

`head()`: show the first six rows of data

`tail()`: show the last six rows of data

`str()`: structure of data

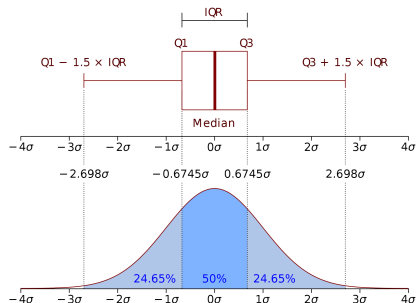
```
head(iris_data)
head(iris_data, n = 10)

tail(iris_data)
tail(iris_data, n = 25)

str(iris_data)
```

Descriptive Statistics

- ▶ `min()`: Minimum
- ▶ `max()`: Maximum
- ▶ `median()`: Median
- ▶ `quantile(,p=)`: Quantile
- ▶ `mean()`: Average
- ▶ `sum()`: Total
- ▶ `var()`: Variance
- ▶ `sd()`: Standard Deviation
- ▶ `summary()`: Overview

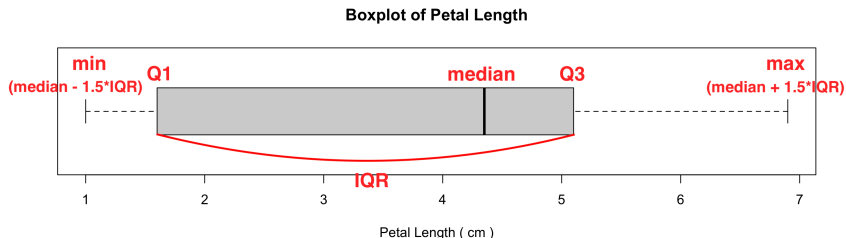


Box Plot

Function: `boxplot(data)`

Example:

```
boxplot(iris_data$Petal.Length,  
        main = "Boxplot of Petal Length",  
        xlab = "Petal Length (cm)",  
        ylab = "",  
        horizontal = TRUE)
```



Getting help

- ▶ `help(mean)`: help about function `mean()`
- ▶ `?mean`: Same. Help about the function `mean()`
- ▶ `example(mean)`: Show an example of function `mean()`
- ▶ `help.search("mean")`: Get help on a specific topic

Practice

- ▶ Read the data "students.csv" into R and name it as "student_data"
- ▶ Compute **minimum**, **maximum**, **mean**, **variance**, and **standard deviation** of math score
- ▶ Draw a **boxplot** of reading score